PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Impact of health literacy and its interventions on health outcomes in those with atrial fibrillation: a systematic review protocol
AUTHORS	Lathlean, Timothy; Kieu, Don; Franke, Kyle; O'Callaghan, Nathan; Boyd, Mark A; Mahajan, Rajiv

VERSION 1 – REVIEW

REVIEWER	Romiti, Giulio Francesco Sapienza University of Rome, Department of Translational and Precision Medicine
REVIEW RETURNED	18-Jul-2022

GENERAL COMMENTS	In this manuscript, the authors report a protocol for a systematic review on health literacy and related-interventions on several types of outcomes in AF patients. The idea underlying this meta-analysis is interesting and this paper may be a good addition to the current body of evidence. There are some limitations/area of uncertainty in this protocol. My comments in detail: - The Introduction paragraph is very long, and may not be easy to understand and to follow - may the authors want to shorten a bit, perhaps trying to present a more straightforward interpretation of the current background and of the need for their proposed systematic review? I can understand the rationale, but some readers may not, and this would clearly help those who are unfamiliar with the topic. - Some sentences are difficult to understand - see page 4 "These dimension are then situated within 3 domains: a a patient within the healthcare system, disease prevention, and health promotion within the wider community" - what this sentence means? I cannot understand. There are other examples. - In the objectives, The aim #1 may be a bit vague; aim #4 mention "correlation" - will the authors assess 'correlation' or 'association'? These terms are not interchangeable and I advice to make this point more clear. - PPI section - "this systematic review will involve only existing reviews" - perhaps the authors referred to "existing papers/studied"? - Inclusion/Exclusion criteria are very vague - while I understand that this may be on purpose (the authors cannot anticipate what type of health literacy interrventions has been described in literature), this also undermine the ability to understand what the authors will actually do. More details are needed in this section. Also, a specific remark to mobile Health interventions may be

	appropriate, given the increase awareness/use of this methodologies. - The authors reported that gray literature will be searched - may the authors want to specify how? Similarly, Data Synthesis, "If a borad consensus is reached for meta-analysis grouping..." this paragraph is unclear, and there is no details on the actual process that will be followed to decide whether or not a meta-analysis will be performed. - Page 9, "The Downs and Black's 27 item checklist..." - I think a verb is missing in this sentence. - Can the authors clarify if the data generated for this SR/MA will be disseminated along with the final report/publication? - I was expecting at least a short discussion on the anticipated clinical implications of this work - for example, what this work is expected to provide in terms of novel findings, etc. Some reflections on these aspects are needed.

REVIEWER	Näslund, Ulf Umea Universitet
REVIEW RETURNED	20-Sep-2022

GENERAL COMMENTS	In general a well-written systematic review protocol, which with a few exceptions conforms with PRISMA-P. Comments and questions: 1. In the Introduciton, last sentence of para 2 the problem with definitions of precision medicine is mentioned - right so. However, the problem with definitions of health literacy (HL) and particularly the issue which methods and quesitons/enquiries to be used methodologically to estimate HL is not described in the background and rationale. 2. Page 8, information sources and search strategy: Which years of publications are to be included? No limitations on date, is this really correct? There is a risk to include systematic reviews published before the research filed was mature - not many years ago?. 3. Page 8, search strategy: No limitations on language. There is a great risk to include non-english systematic reviews, particularly to rely on Google Translate. 4. Last sentence page 8: What kind of disagreements will require a third reviewer and how will the third reviewers judgement be handled?

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1

Dr. Giulio Francesco Romiti, Sapienza University of Rome

Comments to the Author:

In this manuscript, the authors report a protocol for a systematic review on health literacy and related-interventions on several types of outcomes in AF patients.

The idea underlying this meta-analysis is interesting and this paper may be a good addition to the current body of evidence.

There are some limitations/area of uncertainty in this protocol. My comments in detail:

- The Introduction paragraph is very long, and may not be easy to understand and to follow - may the authors want to shorten a bit, perhaps trying to present a more straightforward interpretation of the current background and of the need for their proposed systematic review? I can understand the rationale, but some readers may not, and this would clearly help those who are unfamiliar with the topic.

Please see a revised introduction section as follows:

“Atrial fibrillation (AF) is the most common sustained cardiac arrhythmia and represents significant morbidity and mortality worldwide.(3) The worldwide prevalence of AF is estimated at 0.51% of population and projected to increase further.(4) In addition to an increased risk of stroke, heart failure, and all-cause death, AF continues to contribute to the growing burden on the healthcare system and its resources.(3)

Health literacy plays a critical role within a patient’s healthcare and quality of life, and improving it has been a recognised goal by the World Health Organisation (WHO).(19, 20) Health literacy has been defined according to three types: functional health literacy, interactive health literacy and critical health literacy. (18) It involves people’s knowledge and interactions with health information via four dimensions: accessing, understanding, appraising, and applying. (17) Health literacy can be considered as an aspect of ‘precision health’, which involve the processes of lifetime monitoring and collection of health data, analysis of collected data, and subsequent informed, personalised decision making and interventions.(7-10) A recent scoping review (8) identified nine categories of data that can be collected: sociodemographic, anthropometric, clinical, genomic, perspective, behavioural, psychological, nutritional, and socioenvironmental data.

Currently, a range of definitions and terms have been used in the health literacy literature, with different aims and contexts, which has created uncertainty and a lack of clarity as to what these constitute. This is particularly pronounced when considering health literacy interventions as a core component of precision health initiatives. Our systematic review and meta-analysis aims to evaluate the effectiveness of health literacy interventions for people with AF. In doing so, we aim to synthesise the current literature to identify the impacts of health literacy interventions on a) health literacy and knowledge; b) patient action and behaviour change arising from health literacy and knowledge and c) health outcomes (e.g. quality of life, hospitalisations and/or morbidity/mortality) as a consequence to action taken arising from improvements in health literacy.”

- Some sentences are difficult to understand - see page 4 "These dimension are then situated within 3 domains: a patient within the healthcare system, disease prevention, and health promotion within the wider community" - what this sentence means? I cannot understand. There are other examples.

Thank you for this comment. For clarity, we have removed the sentence relating to 3 domains as we understand this terminology is unclear and that there are several other examples of where health literacy can be improved.

- In the objectives, The aim #1 may be a bit vague; aim #4 mention "correlation" - will the authors assess 'correlation' or 'association'? These terms are not interchangeable and I advice to make this point more clear.

Thank you for this clarification. We have updated our aims to the following:

1. How do health literacy interventions generate additional benefits for precision health initiatives?
2. What questionnaires and tools are currently available and have been used to assess health literacy interventions generally and in AF?
3. How effective are health literacy interventions in improving both knowledge and subsequent beneficial actions for AF?
4. How effective are health literacy interventions that improve knowledge and subsequent beneficial actions in resulting in improved health outcomes such as quality of life, hospitalisations and reduced morbidity/mortality?

- PPI section - "this systematic review will involve only existing reviews" - perhaps the authors referred to "existing papers/studied"?

Our apologies for this confusion. This should read 'this systematic review will involve only existing primary research papers'. Please see our response to the editor in relation to patient and or public involvement in this research.

- Inclusion/Exclusion criteria are very vague - while I understand that this may be on purpose (the authors cannot anticipate what type of health literacy interrventions has been described in literature), this also undermine the ability to understand what the authors will actually do. More details are needed in this section. Also, a specific remark to mobile Health interventions may be appropriate, given the increase awareness/use of this methodologies.

Please see the inclusion/exclusion criteria updated as follows:

Interventions

This review will consider studies that evaluate interventions and initiatives focused on health literacy that are aimed at improving any of the following 1) knowledge and/or literacy 2) patient action and behaviour change and 3) health outcomes, including quality of life, hospitalisations, morbidity/mortality.

Interventions may endeavour to improve any dimension,(17) typology,(18) or similar aspect of health literacy, such as a patient's ability to read, write, approach numeracy, communicate, and/or use electronic technology.(24)

Interventions considered can address health literacy in any domain(17) or setting, and may be in the form of different medium. This can involve face-to-face mediums, such as verbal communication, written communication, visual aids, and supportive systems.(24, 25). We also acknowledge the growing usage of remote mediums and digital initiatives, such as online resources, web-based resources, social media, and mobile applications.

Such intervention fits within Gambhir et al.'s(7) model of precision health within the "Integrated health portal", allowing for learning and adaptation that feeds back into the rest of the model.

- The authors reported that gray literature will be searched - may the authors want to specify how?

Thank you for this clarification. Gray literature, or evidence not published in commercial publications can include academic papers, such as theses/dissertations, research and/or governmental reports, conference papers. We intend to search for such literature through resources such as OpenGrey, Wide-ranging Online Data for Epidemiologic Research (WONDER), Scopus, Proquest Digital Dissertations and Digital Dissertations and Conference Papers Index (CPI) and Zetoc. We also aim to hand-search through relevant conference proceedings and abstracts such as the Cardiac Society of

Australia and New Zealand (CSANZ), Asia Pacific Heart Rhythm Society (APHRS) and Heart Rhythm Society (HRS) Conferences.

Similarly, Data Synthesis, "If a broad consensus is reached for meta-analysis grouping..." this paragraph is unclear, and there is no details on the actual process that will be followed to decide whether or not a meta-analysis will be performed.

We apologies for this confusion. Please see this section updated as follows:

In the case of substantial heterogeneity between methods and outcomes and/or instances where there are not sufficient number of outcomes that can be analysed (e.g. according to the three questions of 1) knowledge and/or literacy 2) patient action and behaviour change and 3) health outcomes, including quality of life, hospitalisations, morbidity/mortality) we will carry out a best evidence synthesis in the place of meta-analysis.

- Page 9, "The Downs and Black's 27 item checklist..." - I think a verb is missing in this sentence.

Thank you for this edit. We have updated this to:

“The Downs and Black's 27 item checklist (45) can be used to assess methodological quality of randomised and non-randomised trials.”

- Can the authors clarify if the data generated for this SR/MA will be disseminated along with the final report/publication?

“The data generated from this SR/MA such as Downs and Black’s rating of quality and/or risk of bias as well as data extraction will be available as supplementary material along with the final report/publication”.

- I was expecting at least a short discussion on the anticipated clinical implications of this work - for example, what this work is expected to provide in terms of novel findings, etc. Some reflections on these aspects are needed.

Thank you for this point. Please see a short discussion on the clinical implications outlined as follows:

This research aims to comprehensively synthesize through systematic review and meta-analysis the existing published and gray literature of health literacy interventions as a component of precision health to improve a) knowledge and literacy b) patient action and behavior change and c) health outcomes. We anticipate this information being used to inform the assessment and future implementation of health literacy interventions not only improving the use of such terms in research and practice but having impact on patient outcomes, cost-effectiveness and accessibility.

Reviewer: 2

Prof. Ulf Näslund, Umea Universitet

Comments to the Author:

In general a well-written systematic review protocol, which with a few exceptions conforms with PRISMA-P. Comments and questions:

1. **In the Introduciton, last sentence of para 2 the problem with definitions of precision medicine is mentioned - right so. However, the problem with definitions of health literacy (HL) and particularly the issue which methods and quesitons/enquiries to be used methodologically to estimate HL is not described in the background and rationale.**

Thank you for this comment. Please see the introduction updated as per Reviewer 1's comments also. Specifically, the following has been added:

"Currently, a range of definitions and terms have been used in the health literacy literature, with different aims and contexts, which has created uncertainty and a lack of clarity as to what these constitute. This is particularly pronounced when considering health literacy interventions as a core component of precision health initiatives."

2. **Page 8, information sources and search strategy: Which years of publications are to be included?**

No limitations on date, is this really correct? There is a risk to include systematic reviews published before the research filed was mature - not many years ago?.

Thank you for this query. We aim to search relevant databases for published material over the past 10 years.

3. **Page 8, search strategy: No limitations on language. There is a great risk to include non-english systematic reviews, particularly to rely on Google Translate.**

Thank you for this question. We have decided to amend our protocol to include studies only published in English. We understand that this may introduce a source of bias and that we have multi-linguists in our team; however believe this would be most consistent with similar systematic reviews and meta-analyses in this field.

4. Last sentence page 8: What kind of disagreements will require a third reviewer and how will the third reviewers judgement be handled?

Disagreements between reviewers where one reviewer has recommended a study for inclusion based on full-text review whereas the other has not will automatically be escalated in Covidence. Instances where the third adjudicating reviewer decides yes will be included, whereas those stated no will be excluded. In cases where the 3rd reviewer is uncertain these will be discussed as a group.